Name: Reg No: Date:

Part A: Debugging Implementation

1. Design a python program/code module with intentional bugs (runtime, logical, and syntax).

2. Use an IDE to:

- Set breakpoints and watch expressions
- Step through the code
- Inspect variables during execution

3. Fix all bugs using appropriate debugging tools and implement exception handling for runtime errors.

4. Document each bug encountered, how it was identified, and how it was fixed in a debugging report. Include screenshots. (**This is very important**.)

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Part B: Unit Testing with PyTest

1. Write unit tests for your project using PyTest. Ensure coverage of:

- Equivalence Partitioning and Boundary Value Analysis (Black-box)
- Statement, Branch, and Path Coverage (White-box)

2. Ensure test coverage for all functions, including edge cases and error handling.

3. Submit the following:

- Source code files (.py)
- Test files (e.g., test_project.py)
- Screenshots of passing test runs

4. Include a testing summary report explaining the approach, coverage achieved, and challenges faced.

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Submission Checklist

Each group must submit the following:

- All Python source files (.py)
- All test scripts (.py)
- Screenshots of debugging and test executions
- Group Report (Word or PDF) with:
- Project overview
- Team members and task distribution
- Debugging logs and bug-fix reports
- Unit testing strategy and summary

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